Project 1: Minimal Pairs Evaluation of Language Model(s) beyond English
A Hungarian Minimal Pair Suite for Negation-Induced Preverb Inversion

Oszkár Urbán
University of Cambridge
ou222@cam.ac.uk

Abstract

Minimal pair research on Large Language Models (LLMs) has primarily focused on high-resource languages such as English, Chinese or Japanese. This work addresses the gap in Hungarian, by introducing a minimal-pair suite focused on the linguistic phenomena: negation-induced preverb inversion. We evaluate the performance on monolingual and a multilingual language models across a range of four lexical complexity levels, introducing intervening adverbs to test the robustness of non-local dependencies. Our results demonstrate that monolingual models achieve superior performance and perform a detailed error analysis. Through tokenization analysis, we show that performance is bottlenecked by tokenization artifacts. Using mechanistic interpretability, we demonstrate that model failure in intervener setups is driven by attention collapse between the negation particle and the preverb-verb construction.

1 Introduction

Human grammar is modelled as a system of abstract constraints that allow some sentences while ruling out closely related alternatives. Linguistic research in language models approximates this competence by assigning higher probability to acceptable sentences than to minimally different ungrammatical variants. Minimal-pair experiments employ this approach by presenting a model with two nearly identical sentences that differ only in a single linguistic feature. By comparing the probabilities a model assigns to each member of the pair, we can evaluate the model's linguistic knowledge.

The use of minimal pairs to probe language models was expanded into large-scale benchmarks such as BLiMP (English) (Warstadt et al., 2023). These frameworks have since been adapted for several high-resource languages, including CLiMP (Chinese) (Xiang et al., 2021) and JBLiMP (Japanese) (Someya and Oseki, 2023), providing essential insights into how model architecture and training data influence the acquisition of syntax.

However, Hungarian remains significantly underrepresented in this area of research. Existing Hungarian benchmarks, such as HuLU (Ligeti-Nagy et al., 2024) and HuGME (Ligeti-Nagy et al., 2025) focus mainly on high-level natural language understanding, instead of probing syntactic knowledge. Hungarian is an agglutinative language characterized by rich inflectional morphology and a flexible word order. Evaluating such languages is crucial for understanding whether LLMs can generalize grammatical rules beyond the relatively simpler syntactic structures found in high-resource language benchmarks, such as English.

In this paper, we explore preverb inversion under negation, a defining feature of Hungarian syntax. In declarative sentences, certain verbal prefixes (preverbs) typically precede the verb stem (e.g., *elmegy* - "goes [away]"). However, the introduction of a negation particle triggers a mandatory inversion, where the preverb must move to a post-verbal position (e.g., *nem megy el* - "NOT [away] goes").

This phenomenon presents a significant challenge for LLMs, as it requires the model to track dependencies that can become non-local when "interveners" are placed between the negation marker and the verb. The main contribution of our work on Hungarian minimal pairs can be summarized:

- Curation of a Minimal Pair Dataset: We introduce a corpus-derived dataset featuring four levels of complexity: ranging from bare negation, in context use, all the way to sentences containing multiple intervening adverbs to test long-distance dependencies.

- Error Analysis: We evaluate model performance using multiple metrics, including Syntactic Log-Odds Ratio (SLOR) (Kann et al.,

2018), to isolate grammatical failures from lexical or length biases. Beyond reporting global results, we conduct a granular error analysis by categorizing failures according to the lexical categories of the preverbs. This approach allows us to assess the models performance from both a macro- and micro-level.

- Mechanistic Interpretability and Tokenization: We investigate the underlying causes of model failure by analysing tokenization artifacts and attention mechanisms. We identified "attention decay" between negation markers and preverb-verb explaining how interveners impacts the models' internal representations.

Minimal pair dataset as well as code can be found online at [1]

2 Minimal Pair Dataset Generation

To evaluate the model's knowledge of Hungarian syntax, we constructed a targeted minimal-pair dataset derived from the Universal Dependencies (UD) (Vincze et al., 2017) to ensure linguistic authenticity.

2.1 Phenomenon: Negation-induced Preverb Inversion

Hungarian verbs frequently combine with verbal prefixes, or preverbs (e.g., *el-*, *ki-*, *meg-*), which encode aspectual or directional meanings. In neutral affirmative sentences, preverbs appear as prefixes immediately preceding the verb. For instance, in a neutral clause, the preverb-verb construction *elmegy* ("goes away") or *kimegy* ("goes out") forms a single linguistic unit.

In negative clauses, however, the negative particle *nem* (or the prohibitive *ne*) triggers a mandatory syntactic inversion. The particle forces the preverb to detach from the verb and move to a post-verbal position. This results in contrast between grammatically sound negated sentences like (1,2,3a), where *nem* directly precedes the verb, and ungramatical variants (1,2,3b), where the preverb remains attached to the verb, as shown in Table 1.

Minimal pair examples	
(1a)	Nem megy el. NEG go.3SG PV.away 'Is not going away.'
(1b)	*Nem elmegy. NEG PV.away-go.3SG
(2a)	Nem derült ki. NEG turn_out.3SG.PST PV.out '(It) did not turn out.'
(2b)	*Nem kiderült. NEG PV.out-turn out.3SG
(3a)	Nem lép ki. NEG step.3SG PV.out 'Do not step out.'
(3b)	*Nem kilép. NEG PV.out-step.3SG

Table 1: Minimal pair examples for negation-induced preverb inversion. (a) denotes the grammatical form; (b) denotes the ungrammatical variant. Glosses: 3SG: 3rd person singular, NEG: negation, PV: preverb.

2.2 Source Data

To ensure a diverse and linguistically correct samples, we utilized the UD_Hungarian-Szeged corpus (including training, development, and test sets). We identified candidate preverb-verb pairs by locating tokens labelled with the compound:preverb dependency relation.

From this pool, the dataset generator sampled unique verb–preverb occurrences. For each pair, we selected the appropriate negation operator based on the verb's mood feature in the UD metadata. The prohibitive *ne* was assigned for imperative or subjunctive moods, while the declarative *nem* was used for all other instances. Initially, we identified 413 unique preverb–verb pairs. However, to address data sparsity where some preverbs appeared with only a few verbs, we manually expanded the dataset. By ensuring each preverb occurs with at least 10 unique verbs, the final dataset was increased to 612 unique preverb–verb minimal pairs.

2.3 Interveners

To test the model's syntactic robustness we expanded the base verbal pairs into a hierarchy of four experimental conditions representing increasing levels of lexical difficulty:

1. Bare: This condition presents the introduced phenomena in isolation.

2. Context: In this condition, both the grammatical and ungrammatical example remain unchanged compared to Bare, but the bare phenomena is put in a sentence. The aim of this experiment is to better reflect natural language usage.

[1] https://github.com/oszkarurban/l95-minimalpair

3. Intervener 1: A single adverb (e.g., *már* “already”, *soha* “never”) is inserted between the negation particle and the finite verb. This forces the model to track the inversion requirement across distance.

4. Intervener 2: A second adverbial phrase is inserted, yielding longer clauses where the negation–verb dependency spans two interveners. This serves as a stress test for long-distance dependency.

Table 2 shows a minimal pair example over all four experiment conditions.

Condition	Example (Pair: ki + lép)
1. Bare	Nem lép ki. *Nem kilép. *(Not steps [out] / *Not [out]-steps)*
2. Context	Nem lép ki a nagy szobából. *Nem kilép a nagy szobából. *(Not steps [out] from the big room)*
3. Intervener 1	Nem aztán lép ki a nagy szobából. *Nem aztán kilép a nagy szobából. *(Not then steps [out] from the big room)*
4. Intervener 2	Nem aztán pont lép ki a nagy szobából. *Nem aztán pont kilép a nagy szobából. *(Not exactly then steps [out] from the big room)*

Table 2: Example minimal pair for the Hungarian verb-particle construction *ki* + *lép* ("steps out") across four experimental conditions: Bare, Context, Intervener 1, and Intervener 2. Each entry includes the Hungarian text followed by its English translation.

2.4 Linguistic Classification of Preverbs

An important aspect in our experiment is the varying semantic meaning of the preverbs. While all preverbs follow the same syntactic inversion rule, their lexical role varies. To investigate whether semantic meaning of preverbs affects a model's syntactic robustness, we categorized the preverbs in our dataset into three distinct groups based on their linguistic function, according to (Kiss, 2002):

- Aspectual Preverbs: These preverbs, such as *meg*, *el*, and *újra*, lack concrete lexical meaning on their own.

- Spatial/Directional Preverbs: This group includes preverbs with clear, independent meaning, such as *ki* ("out"), *be* ("in"), *fel* ("up"), and *le* ("down"). These preverbs have significant semantic meaning and provide a concrete directional component to the verbal predicate.

- Figurative Preverbs: These are preverbs that form idiomatic units with specific verbs (e.g., *létre* + *hoz* = "to create"; *észre* + *vesz* = "to notice"). In these cases, the preverb often has no independent meaning outside of the specific verbal construction.

3 Experimental setup

3.1 Model selection

We select two models, Goldfish goldfish-models/hun_latn_1000mb (Chang et al., 2024) and BabyLM-community/ALL-baseline-large (Jumelet et al., 2025a), which share a common Transformer-based architecture but differ in their training approaches and training data. We use this comparison to determine which approach better facilitates the acquisition of our syntactic phenomenon.

Monolingual vs. Multilingual By comparing the monolingual Goldfish-1000mb with the multilingual BabyLM-ALL-baseline-large, we test whether syntactic knowledge requires exclusive exposure to a single language or if it can emerge effectively from a shared multilingual space. This comparison allows us to investigate whether cross-linguistic transfer in multilingual models hinders the learning of syntactic rules.

Training data Goldfish-1000mb is trained on broad web-based corpora typical of traditional LLM pre-training. In contrast, BabyLM uses a curated dataset designed to mimic how children acquire language. This comparison is motivated by the ongoing research regarding the advantage of Child-Directed Language in acquiring syntactic abilities, as proposed by (Huebner et al., 2021) but challenged by (Padovani et al., 2025).

3.2 Metrics

We utilized three distinct metrics:

- Negative Log-Likelihood (NLL): Following the MultiBLiMP (Jumelet et al., 2025b) approach, we calculate the difference between the NLL of the ungrammatical and grammatical sentences. A positive delta indicates that the model assigns a higher probability to the grammatical member.

- Length-Normalized Log-Probability (LP-Norm): Because Hungarian preverb inversion results in changes splitting the preverb from the verb that alter token counts, we normalize the total log-probability by the sequence length to prevent a bias toward shorter forms.

- Syntactic Log-Odds Ratio (SLOR): SLOR serves as our most important metric. It extends the length-normalization of LP-Norm by further adjusting the model's scores against the natural unigram frequency of the tokens. By "penalizing" the probability of common words, SLOR ensures the model is rewarded for recognizing the syntactic arrangement rather than merely identifying high-frequency lexical sequences. The SLOR metric requires a unigram probability map. We constructed this using Webcorpus 2.0 (Nemeskey, 2020), a 9-billion-word dataset of Hungarian web text.

4 Experiments

4.1 Global performance

In this section, we analyze the global performance of the monolingual Goldfish and multilingual BabyLM models across our four experiment variations (Bare, Context, Intervener, Intervener2). We report accuracy as the percentage of cases where the model correctly identifies the grammatical sentence in a minimal pair choice.

Comparing results between models The most important finding is that the monolingual Goldfish model significantly outperforms the multilingual BabyLM on every metric for all four experiment setups. This supports the hypothesis that isolated exposure to Hungarian allows a model to learn the linguistic knowledge necessary for the phenomena more effectively than a multilingual model.

Goldfish model The Goldfish model demonstrates high performance in the Bare setting, where linguistic phenomenon is presented in isolation, achieving 93.30% NLL and 79.74% SLOR accuracy. These results remain comparably strong in the Context experiment (92.97% NLL, 83.66% SLOR), indicating robust understanding of the phenomena.

However, performance declines with interveners. In the first Intervener setting, the SLOR accuracy drops to 73.69%, and the decline is most pronounced in Intervener 2, where accuracy falls to 52.45%. This downward trend suggests that the model's performance is sensitive to the distance between negation marker and preverb-verb construction. This is consistent with Hu et al. (2020), who demonstrated that the syntactic generalization of Transformers is fragile when dependencies are stretched by intervening distractors.

We can see that reporting multiple metrics across experiments is crucial, particularly since sequence lengths grow substantially from Bare to Intervener conditions, making length-normalized scores like LP and SLOR essential for accurate assessment. In the Intervener 2 condition, while raw NLL accuracy drops to 47.39%, LP and SLOR accuracies remain considerably higher at 60.78% and 52.45%, respectively. This suggests that the observed degradation is due to penalty for the increased number of tokens, rather than a complete loss of the model's syntactic knowledge.

BabyLM model The BabyLM model follows a similar downward trajectory as experiment complexity increases. While it achieves $\approx$ 58% accuracy in the Bare condition, its performance drops to near-random $\approx$ 51–54% in Context and Intervener experiments. These results suggest that multilingual training alone is insufficient for the model to robustly acquire the preverb inversion phenomena.

Probability Margin Beyond accuracy in selecting the grammatical option, we calculate the model's SLOR Margin, defined as the difference in SLOR scores between the grammatical and ungrammatical options.

As experiment complexity increases, the mean SLOR Margin significantly decreases across versions. The margin drops from 1.188 (Bare) and 0.599 (Context) to 0.367 (Intervener) and 0.024 (Intervener 2). A similar trend is observed in the BabyLM results, where the margin narrows from 0.233 (Bare) to 0.047 (Context), 0.032 (Intervener), and 0.042 (Intervener 2).

This narrowing margin indicates that increased lexical task complexity reduces the model's discriminative certainty between grammatical and ungrammatical version.

4.2 Preverb Categorization

As described in Subsection 2.4 we categorize the preverbs based on their linguistic properties: Aspectual Preverbs, Spatial/Directional Preverbs, and Figurative Preverbs. The purpose of this classification is to analyse whether minimal pair accuracy vary across these groups.

	Goldfish			BabyLM		
Version	**NLL (%)**	**LP-N (%)**	**SLOR (%)**	**NLL (%)**	**LP-N (%)**	**SLOR (%)**
Bare	93.30	82.84	79.74	57.03	58.33	58.82
Context	92.97	85.46	83.66	51.96	52.45	52.12
Intervener	79.41	79.25	73.69	51.47	53.43	51.14
Intervener2	47.39	60.78	52.45	51.96	53.76	54.08

Table 3: Accuracy of Goldfish and BabyLM on our minimal pair datset, using NLL, LP-Norm, and SLOR. The results show that the monolingual Goldfish model consistently outperforms the multilingual BabyLM across all experiments. A performance degradation is observed for both models as the experiment complexity increases from the Bare condition to Intervener 2.

As shown in Figure 1 the general trend is that the model performs best on Aspectual preverbs before Figurative and lastly Spatial/Directional.

A plausible explanation for the higher performance of Aspectual Preverbs is that they lack independent semantic meaning and instead require a specific, "fixed" preverb-verb coupling. Because these combinations are grammatically mandatory, the model learns them as strict rules. In contrast, Spatial/Directional preverb-verb combinations are less strict and more interchangeable (e.g., many different spatial preverbs can logically pair with the same motion verb). A plausible explanation is that this flexibility results in a looser coupling.

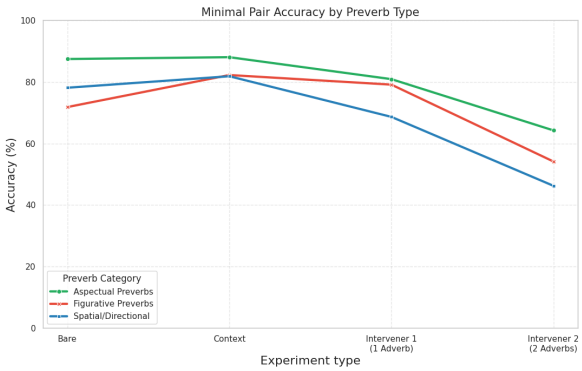

Figure 1: Model accuracy broken down by preverb category (Aspectual, Figurative, and Spatial/Directional). The results indicate that the model performs best on Aspectual Preverbs, likely due to their fixed grammatical coupling, whereas performance is lowest for the Spatial/Directional Preverbs likely due to more flexible preverb-verb pairings due to their independent semantic meaning.

4.3 Impact of tokenization

In this section, we investigate whether the model's performance on the introduced phenomena is determined by how the tokenizer represents the preverb-verb construction. We define "well-split" cases where preverb and verb are treated as separate tokens, or are tokenized in a way that respects the boundary between the preverb and the verb.

	Goldfish		BabyLM	
Category	**N**	**SLOR (%)**	**N**	**SLOR (%)**
Well-Split	300	97.67	229	70.31
Rest	312	62.50	383	51.96

Table 4: Accuracy comparison between Goldfish and BabyLM across tokenization categories for Base experiment. "Well-split" denotes cases where the tokenizer respects the preverb-verb boundary, while "Rest" includes all other instances. The significantly higher performance in "Well-split" cases highlights that boundary-aware tokenization is a key for our linguistic phenomena. N indicates the number of examples in that category.

Goldfish As shown in Table 4, in the "well-split" cases the model achieves near perfect accuracy 97.67%, and the accuracy drops significantly to 62.50% in cases where the preverb-verb boundary is ignored. These results suggest that for understanding the phenomenon it is crucial that we need a tokenizer that respects the structural boundary between the preverb and the verb, so that it can break the construction apart or reorder it during negation.

BabyLM Although to a lesser extent, we similarly observe that well-split tokenization leads to higher performance in the BabyLM model as well, 70.31% in well-split compared to 51.96% in rest of the cases. The fact that BabyLM still lags behind Goldfish in the "well-split" category suggests that the shared multilingual vocabulary hinder the language-specific knowledge for this phenomenon, in line with (Rust et al., 2021).

Table 5: Goldfish model. Examples of correctly identified minimal pairs. We can observe that the tokenizer represents the preverb-verb construction as separate tokens in grammatical cases, whereas it is represented as a single, atomic token in ungrammatical cases. Preverbs are highlighted in bold.

Pair: meg + valósít Ungramm: ['_Nem', '_megvalósít'] Gramm: ['_Nem', '_valósít', **'_meg'**]
Pair: ki + derült Ungramm: ['_Nem', '_kiderült'] Gramm: ['_Nem', '_derült', **'_ki'**]
Pair: ki + lép Ungramm: ['_Nem', '_kilép'] Gramm: ['_Nem', '_lép', **'_ki'**]

Table 6: BabyLM model. Examples of incorrectly identified minimal pairs. In these instances, the tokenizer ignores the preverb-verb boundary in ungrammatical cases, leading to an "overflow" where the preverb is merged into the subsequent tokens. Preverbs are highlighted in bold

Pair: vissza + keresi Ungramm: ['Nem', **'v', 'iss', 'zak'**, 'er', 'esi'] Gramm: ['Nem', 'ker', 'esi', **'v', 'iss', 'za'**]
Pair: szembe + nézhetnek Ungramm: ['Nem', **'sz', 'emb', 'en'**, 'e', 'zh', 'et', 'n', 'ek'] Gramm: ['Nem', 'ne', 'zh', 'et', 'n', 'ek', **'sz', 'emb', 'e'**]
Pair: hátra + helyezi Ungramm: ['Nem', **'ha', 'tr'**, 'ah', 'el', 'yez', 'i'] Gramm: ['Nem', 'hel', 'yez', 'i', **'ha', 'tra'**]

4.4 Attention Mechanism

In this section, we apply mechanistic interpretability techniques to analyze the internal attention patterns. Given the results in previous section we will selectively focus on Goldfish-1000mb monolingual model as the multilingual BabyLM achieved near random accuracy.

- The Source: The negation marker (*nem* or *ne*).
- The Critical Region: The preverb-verb construction, consisting of the verb stem and its associated preverb.

Following (Clark et al., 2019), rather than measuring attention across the entire sequence, we focused on the information flow between two linguistic units:

For every sentence in our dataset, we identified the tokens belonging to the critical region, accounting for subword splits and tokenizer artifacts. We then applied Max-Head Pooling to capture the strongest signal across all attention heads in each layer. The resulting metric represents the total "weight" the preverb-verb construction assigns to the negation marker, that we refer to as Attention Mass.

Grammatical vs. Ungrammatical Attention Pattern As shown in Figure 2, there is a divergence in Attention Mass between grammatical and ungrammatical sentences. In grammatical sentences, the tokens in the critical region maintain a higher attention weight back to the negation marker compared to ungrammatical sentences. This indicates that the model's preference for the correct, is tied to the strength of this attention. This effect is most prominent in the early-to-middle layers, specifically layers 4-5.

Attention Decay To understand why model performance collapses in more complex experiments, we analysed the Attention Mass across increasing experiment difficulty, comparing Bare and Intervener 2.

The data reveals a significant decay of the attention signal as intervener distance increases. Comparing the *Bare* condition to

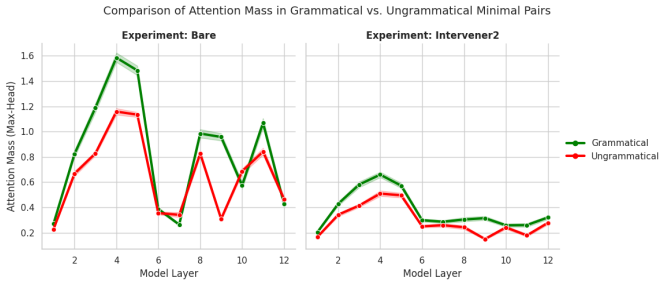

Figure 2: Attention analysis for the Goldfish-1000mb model. Grammatical sequences maintain higher attention mass from the preverb-verb construction to the negation marker, especially noticeable in early-mid layers. Interveners cause a decline in attention mass, correlating with the model's performance drop. The shaded areas represent the 95% confidence interval.

the *Intervener 2* condition, where two adverbs separate the negation particle from the target, attention mass for both grammatical and ungrammatical sentences drops significantly. The performance drop on long-distance dependencies is consistent with this attention decay, suggesting that attention pattern might be a plausible explanation.

5 Limitations and Future Work

The main two limitations of our work are related to Model Size and Intervener experiment setup:

Model Size Our evaluation is restricted to the Goldfish-1000mb and BabyLM model. These models are small in parameter count compared to state of the art LLMs. While our work provide valuable findings from a computational linguistic standpoint, future work should investigate whether the findings persists in state-of-the-art LLMs as model as scale increases.

Furthermore, our multilingual setup (BabyLM) was trained on a dataset motivated by how children learn language. Future work should investigate whether the limited performance observed in the multilingual approach persists when models are trained on large-scale general multilingual corpora. This would clarify whether the limited performance is a limitation of multilingual models or a data limitation due to child-directed learning.

Intervener Experiment setup In our intervener experiments, we restricted the distance between the negation particle and the verb to one or two adverbs. While this setup may appear limited from a computational perspective, it was a deliberate choice to ensure linguistic plausibility. In Hungarian adding three or more intervening (e.g., *Nem [intervener 1] [intervener 2] [intervener 3] [intervener 4] ... megy el*) results in sentences that are unnatural. This is the reason why we prioritized linguistically plausible experiment over scaling the intervener sequence lengths.

Furthermore, we focused exclusively on using adverbs as interveners. Future research should expand the variety of interveners rather than just the distance.

6 Conclusion

In this work, we introduced a minimal-pair dataset to evaluate the linguistic performance of language models on Hungarian language. We investigated a syntactic phenomenon called "negation-induced preverb inversion," where the negative particle triggers a mandatory displacement of verbal prefixes. The significance of this phenomenon lies in the fact that negation is a core logical operator; therefore, a model's inability to resolve these dependencies could lead to an inversion of meaning between affirmative and negative statements.

We performed a comparison between the monolingual Goldfish-1000mb and the multilingual BabyLM model. This selection was motivated by a comparison of monolingual versus multilingual architectures, as well as the impact of training on broad web-based corpora versus curated datasets modeled on child language acquisition. Our experiments spanned four levels of increasing syntactic complexity. Ranging from the phenomenon in isolation to "stress tests" involving multiple interveners. We performed both global and granular error analysis and investigated potential explanations, such as tokenization. Our findings reveal:

- **Monolingual Superiority:** The monolingual model significantly outperforms the multilingual baseline, which performs near random, suggesting that multilingual representations hinders the acquisition of syntactic knowledge.
- **Intervener Distance:** Model performance depends on the intervener sequence distance; accuracy collapses as the number of "interveners" between the negation marker and the verb increases.
- **Tokenization** Model performance is significantly higher in cases where the tokenizer treats the preverb and verb as distinct tokens or maintains the boundary between the preverb and the verb. This suggests that tokenization respecting morpheme boundaries is a essential for resolving preverb-verb inversion.
- **Attention:** Using mechanistic interpretability, we identified that the attention mass from the preverb-verb construction to the negation marker is greater in grammatical than ungrammatical sentences. Furthermore, this attention mass dramatically decreases as interveners are introduced, directly mirroring the observed collapse in model accuracy.

Our minimal pair dataset as well as code can be found online at: https://github.com/oszkarurban/l95-minimalpair

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